

M3 Figure 10 in M3 paper: Actually have $y = (g_\chi g_\mu)^2 (m_\chi/m_{Z'})^4 = g_\mu^2 (1/3)^4 = g_\mu^2/3^4$

$$\text{BR}(Z' \rightarrow \text{inv}) = \frac{\Gamma(Z' \rightarrow \nu_\mu, \nu_\tau, \chi)}{\Gamma(Z' \rightarrow \nu_\mu, \nu_\tau, \chi, Z')} = \frac{\sum_{i=\nu_\mu, \nu_\tau, \chi} g_i^2 \left(1 + \frac{2m_i^2}{m_{Z'}^2}\right) \sqrt{1 - \frac{4m_i^2}{m_{Z'}^2}}}{\sum_{i=\nu_\mu, \nu_\tau, \chi, Z'} g_i^2 \left(1 + \frac{2m_i^2}{m_{Z'}^2}\right) \sqrt{1 - \frac{4m_i^2}{m_{Z'}^2}}}$$

Robert (?): Experiment actually tests $g_\mu^2 \text{BR}(Z' \rightarrow \text{inv}) \Rightarrow$ Get bounds for other values of $m_{Z'}, m_\chi, g_\chi$ by using

$$g_{\text{new}}^2 = g_{\text{old}}^2 \times \frac{\text{BR}(Z' \rightarrow \text{inv})_{\text{old}}}{\text{BR}(Z' \rightarrow \text{inv})_{\text{new}}}$$